IBE; discussion

IBE discussion

➔How to impersonate?

- ⇨*Same ID is not sufficient*
- ⇨*Must «register» to PKG to get private key*

➔PKG: very powerful entity

- ⇨*Solution: Distributed PKG!*
  - ➔Use pedersen's DKG
- ⇨*Intermediate approaches: certificateless crypto*
  - ➔See next slide

➔Multi-authority?

- ⇨*Yes, several extensions*

ECQV

➔**Elliptic Curve Qu-Vanstone**

- ⇨*A.k.a. implicit certificates*
- ⇨*Identity-based*

➔**Idea: self-certifying EC point**

- ⇨*Do not use PKG but use a CA*
- ⇨*But CA does* **not** *issue standard certificates*
- ⇨*it «helps» to extract the user public Key from an «implicit» identity-based certificate*

➔**Advantage: very small (implicit) certificates!!**

- ⇨*A single EC group point!*

Initial registration

➔**CA:**

⇨*Private key* ➔ *c*

⇨*Public key* ➔ *C = cG*

Before asking for a certificate, the device I_A generates a random positive integer r_A; then, it computes a point R_A on the chosen elliptic curve, $R_A = r_A \cdot G$, and sends it to the CA. The CA extracts a random positive integer k, computes $k \cdot G$ (another point on the elliptic curve), generates the implicit certificate,P_A, and the implicit signature, γ_A, by using:

$$\begin{cases} P_A = R_A + k \cdot G \\ \gamma_A = c + k \cdot H(P_A, I_A) \end{cases} \quad (1)$$

Now, it sends them to the device. Then, it can generate its private key, $s_{v,A}$, and its public key, $P_{b,A}$, by:

$$\begin{cases} s_{v,A} = \gamma_A + r_A \cdot H(P_A, I_A) \\ P_{b,A} = s_{v,A} \cdot G \end{cases} \quad (2)$$

verification

As well-known, the most important strength of ECQV is that the public key of a given device can be computed by any other third-party starting from the knowledge of the implicit certificate and the public key of the authority:

$$P_{b,A} = s_{v,A} \cdot G = C + P_A \cdot H(P_A, I_A). \qquad (3)$$

➔ **In conclusion:**

- ⇨ *Unlike IBE PKG, no third party knows user private keys*

➔ **Certificates: reduced to single (!) group point!!**

- ⇨ *Very low transport overhead compared to traditional certificates*
  - → O(40) bytes versus o(1000+) bytes

➔ **Application in low-resources IoT/wireless networks**

- ⇨ *See e.g., Sciancalepore, Capossele, Piro, Boggia, Bianchi, **Key management protocol with implicit certificates for IoT systems**. 2015 Workshop on IoT challenges in Mobile and Industrial Systems (pp. 37-42).*
- ⇨ *Used (at that time) only in ZigBee:* **Certificate Based Key Establishment (CBKE) protocol**